

Does integrating multiple sources of information correct detrimental host behavioral responses to perceived disease risk?

Dale Clement, Robert Holt, Nick Ruktanonchai, Omar Saucedo, Nicholas Kortessis

EEID 2026 Poster Handout

Movement Model. Consider N individuals divided among 3 locations: “home”, where transmission is absent ($\beta_{home} = 0$), and low (L) and high (H) risk “away” locations ($0 < \beta_L < \beta_H$). Individuals take “simple” trips from home to another location, remain for a period of time, and then return home. We set $N = 1$ such that the number of individuals in location i , N_i , is a proportion. Individuals visit away location i at a rate $\phi_i = \sigma\nu_i$, where σ is the rate individuals leave home and ν_i is the probability of visiting location i . Individuals return from location i at a rate τ_i . This yields the simple trips movement model

$$\begin{aligned}\frac{dN_i}{dt} &= \phi_i N_{home} - \tau_i N_i \\ \frac{dN_{home}}{dt} &= -(\phi_L + \phi_H)N_{home} + \tau_L N_L + \tau_H N_H.\end{aligned}$$

If ϕ_i^0 is the visitation rate in absence of behavioral response to disease risk, the risk-free equilibrium densities are $N_{home}^0 = \frac{1}{1+\Sigma_i \phi_i^0/\tau_i}$ and $N_i^0 = \frac{\phi_i^0/\tau_i}{1+\Sigma_i \phi_i^0/\tau_i}$. These densities are also used as the initial host densities upon introduction of the disease, hence the naught superscript.

Behavioral responses to disease. Individuals respond to disease by adjusting their visitations rates ϕ_i from the risk-free rates ϕ_i^0 based on perceived risk X_i . In this study, we examine “relative risk responses”: the dependence of visitation probabilities ν_i on the deviation of perceived risk at location i from the average, $X_i - \bar{X}$:

$$\phi_i = \sigma^0 \nu_i^0 \frac{\exp(-c(X_i - \bar{X}))}{\Sigma_i \nu_i^0 \exp(-c(X_i - \bar{X}))}.$$

where c controls the strength of response. The change in ν_i with X_i is standardized so that $\Sigma_i \nu_i = 1$. The equilibrium proportion of individuals at home stays the same for relative risk responses.

Disease Model. We consider a susceptible-infected-susceptible (SIS) compartmental disease model coupled to the simple trips movement model. Because the base movement model is written in terms of N_i , we track changes in N_i in place of changes in S_i . We may write the Simple Trips SIS model as

$$\begin{aligned}\frac{dI_i}{dt} &= \lambda_i(N_i - I_i) - \gamma I_i + \phi_i I_{home} - \tau_i I_i \\ \frac{dI_{home}}{dt} &= -\gamma I_{home} - (\phi_H + \phi_L)I_{home} + \tau_L I_L + \tau_H I_H.\end{aligned}$$

The force of infection, λ_i is given by the equation

$$\lambda_i = \beta_i \frac{I_i}{(1-p)N_i + p}$$

where the parameter p controls the dependence of transmission on host density. When $p = 0$, transmission is purely frequency-dependent. When $p = 1$, transmission is purely density-dependent.

We calculated the “optimum that minimizes prevalence”, the dashed line in the post results figures, as the initial proportion of individuals in the low risk patch that results in the lowest equilibrium global prevalence I^*/N when there is no behavioral response to disease risk (i.e., $\min_{N_L^0 \in (0,1)} \{I^* | c = 0\}$). We did not find multiple optima over the range of model parameters we examined.

Risk Proxies

We construct each risk proxy as a combination of three underlying types of information: disease prevalence $\frac{I_j}{N_j}$, host density N_j , and transmission coefficients β_j . Because disease-dependent behavior requires risk proxies to be a function of infections, we assume that each risk proxy contains two components: the information signal(s) about local risk and information about global disease prevalence I . The three risk proxies examined in our previous study were prevalence-based risk perception ($X_j = \frac{I_j}{N_j}$), density-based risk perception ($X_j = N_j I$), and place-based risk perception ($X_j = \beta_j I$).

In the present study, we consider risk proxies constructed from combinations of local prevalence, density, and transmission coefficients. We model risk proxies as linear combinations, on the log scale, of different information types, so that the general form for a risk proxy that combines two or more information types is:

$$X_i = (W_i)^{a_1} (Y_i)^{a_2} (Z_i)^{a_3} I$$

where W_i , Y_i , Z_i are generic placeholders for type of information and the weight placed on a particular information type is given by one of a_1 , a_2 , a_3 . We then scale the combination of local information by global prevalence. Note that in the case of prevalence information, this requires "factoring out" global prevalence from local prevalence (i.e., $\frac{I_i}{IN_i}$ rather than $\frac{I_i}{N_i}$) to avoid double counting global prevalence in the final risk proxy.

An advantage of this log-additive construction, as compared to a purely additive construction, is that when the weights on each piece of information are set to 1, these combinations become other, easily recognizable epidemiological quantities. Combining density and transmission coefficients yields weighted density ($X_j = \beta_j N_j I$), combining prevalence and density yields infection number ($X_j = (\frac{I_j}{N_j}) N_j = I_j$), combining prevalence and transmission coefficients yields the force of infection under frequency-dependent transmission ($X_j = \beta_j \frac{I_j}{N_j}$), and combining all three yields the force of infection under density-dependent transmission ($X_j = \beta_j I_j$).